

EXP. NO: 7

Data Visualization Using Tableau

AIM:

To understand the Tableau interface and perform data visualization by connecting to a CSV data source. The experiment includes creating bar, line, and pie charts, adding a calculated field, and building an interactive dashboard and story.

PROCEDURE:

1. Launch Tableau & Explore Interface

- i. Open Tableau Desktop/Public.
- ii. Explore the Start Page, Data Pane, Worksheet, Dashboard, and Story options.
- iii. Identify the Columns shelf, Rows shelf, Marks card, Filters, and Show Me panel.

2. Connect to Data

- i. Click Connect → Text File.
- ii. Select the Tableau_SalesData.csv file.
- iii. Verify the fields such as Date, Region, Category, Sales, Profit, Quantity, and Target.

3. Create Basic Visualizations

- i. Create a Bar Chart by placing Category on Columns and Sales on Rows.
- ii. Create a Line Chart by placing Date on Columns and Sales on Rows.
- iii. Create a Pie Chart using Region and Sales.

4. Create a Calculated Field

- i. Select Analysis → Create Calculated Field.
- ii. Create the calculated field Profit Margin using: $\text{SUM}([\text{Profit}]) / \text{SUM}([\text{Sales}]) * 100$
- iii. Create a bar chart showing Profit Margin by Category.

5. Build Interactive Dashboard

- i. Create a new dashboard.
- ii. Add the Sales by Category, Monthly Sales Trend, Sales by Region, and Profit Margin by Category charts.
- iii. Add a Region filter and apply it to worksheets using the same data source.
- iv. Arrange and resize the charts to form a 2×2 dashboard layout.

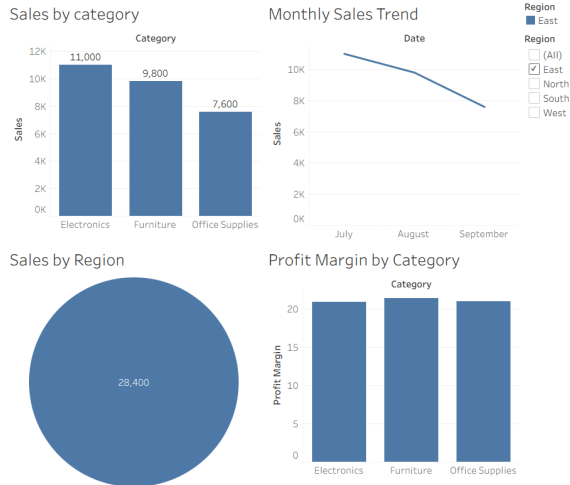
6. Create Story

- i. Create a new Story.
- ii. Add the **Sales Analytics Dashboard** to the story.
- iii. Set the story title as **Overall Sales Analysis**.

OUTPUT:

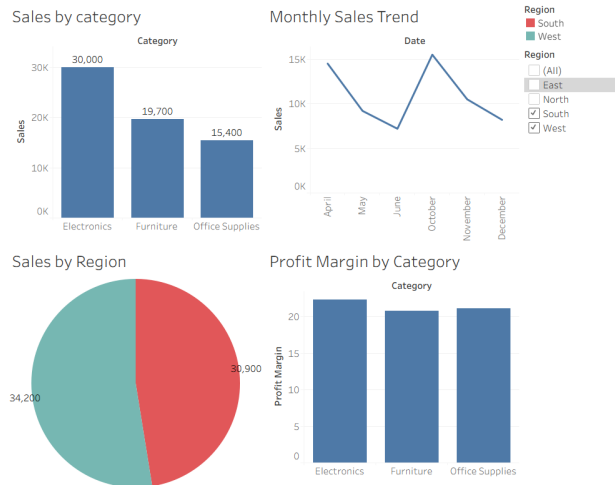
Overall Sales Analysis

Interactive Sales Analysis Dashboard showing sales trends by category, region, and month, along with profit margin analysis.



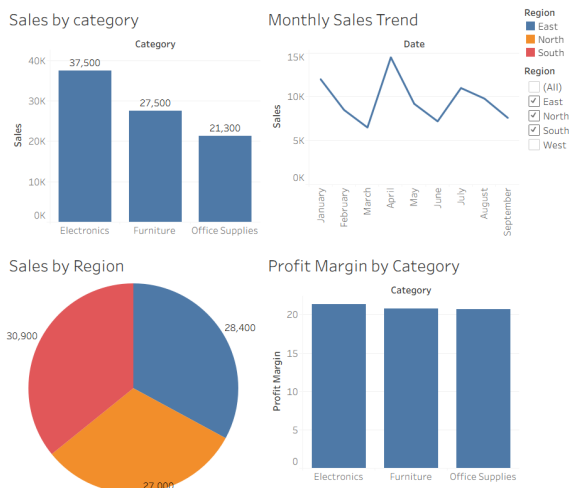
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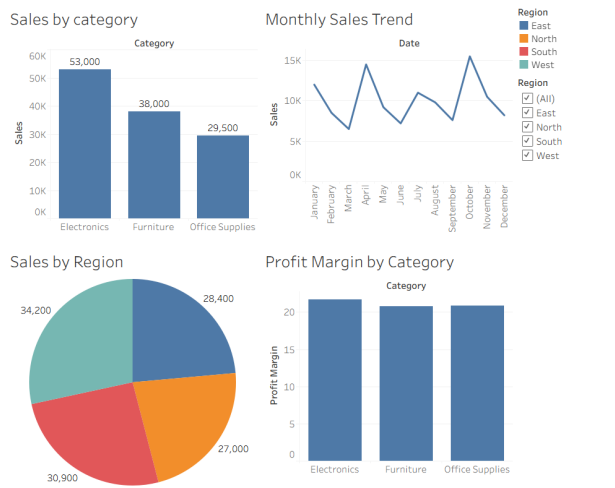
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Interactive Sales Analysis Dashboard showing sales trends by category, region, and month, along with profit margin analysis.



RESULT:

Thus, the given data was successfully connected and visualized using Tableau. Bar, line, and pie charts were created along with a calculated field, interactive dashboard, and story successfully.